Question

How large must a reported mAP gain be before a single run can be trusted?

1

How large are published gains?

Can they be checked?

- 131 papers, 182 evaluations
- 17 reporting items
- claimed gains extracted

2

How much do scores move on their own?

Nothing changed but seed or split

- 5 detectors × 10 seeds
- 2 datasets
- 10 grouped splits × 3 seeds

3

How much does leakage add?

Near-duplicates across train and test

- registration graph, human-checked
- 10 paired counterfactual splits
- 5 partner doses

4

Do common modules survive a re-test?

Matched weights, matched seeds

- 3 modules
- 10 paired seeds
- same split and training budget

Common yardstick the smallest difference in test mAP@0.5 that one run per model can resolve

Reporting checklist | **Seeds-needed rule** | **Released splits, logs and code**